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I am going to flip a fair coin until it comes up heads. If the first time it comes up heads is on the 1st toss, I will give you \$2. If the first time it comes up heads is on the second toss, I will give you \$4. If the first time it comes up heads is on the 3rd toss, I will give you \$8. And in general, if the first time the coin comes up heads is on the nth toss, I will give you $\$2^n$.

What is the expected utility of playing the game?

We can think about this using the following table:

Outcome	First heads is on toss #1	First heads is on toss #2	First heads is on toss #3	First heads is on toss #4	First heads is on toss #5	.....
Payoff	\$2	\$4	\$8	\$16	\$32	.....
Probability	1/2	1/4	1/8	1/16	1/32	.....

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